

Database Requirements Document

Project: World Cup Prediction Game

Version: 1.0.0

Date: 2026-06-11

Status: Draft

Table of Contents

1. [Overview](#)
 2. [Connection Configuration](#)
 3. [Entity Relationship Diagram](#)
 4. [Table Definitions](#)
 5. [Relationships](#)
 6. [Indexes](#)
 7. [Constraints & Business Rules](#)
 8. [Migration Strategy](#)
 9. [Seed Data](#)
 10. [Backup & Maintenance](#)
-

Overview

The database stores all persistent data for the prediction game: admin credentials, match fixtures, participant records, and predictions. Scoring is derived from the `predictions` and `matches` tables — no separate points column is stored, keeping the source of truth in one place.

Engine: MySQL 8.0+

Character Set: `utf8mb4`

Collation: `utf8mb4_unicode_ci`

Tables: 4

Relationships: 2 foreign keys (both on `predictions`)

Connection Configuration

Environment Variables

All connection values must be set in `.env` (never committed). Use `.env.example` as the template.

```
DB_HOST=127.0.0.1
DB_PORT=3306
DB_NAME=prediction_game
DB_USER=pg_app
DB_PASSWORD=<strong-password>
DB_POOL_MIN=2
DB_POOL_MAX=10
DB_CONNECT_TIMEOUT=10000
```

Connection Pool Settings

Parameter	Value	Notes
host	DB_HOST	Use 127.0.0.1, not localhost, to force TCP
port	DB_PORT (default 3306)	
database	DB_NAME	
user	DB_USER	Least-privilege app user (see below)
password	DB_PASSWORD	
connectionLimit	DB_POOL_MAX (default 10)	
waitForConnections	true	Queue requests when pool is full
connectTimeout	10000 ms	Fail fast on misconfiguration
charset	utf8mb4	Required for emoji/unicode support

Least-Privilege App User

The application connects as a restricted user. Run once on the database server to provision:

```
CREATE USER 'pg_app'@'%' IDENTIFIED BY '<strong-password>';

GRANT SELECT, INSERT, UPDATE, DELETE
  ON prediction_game.*
  TO 'pg_app'@'%';

-- Migrations run as a separate, more privileged user (not the app user)
CREATE USER 'pg_migrate'@'localhost' IDENTIFIED BY '<migration-password>';
GRANT ALL PRIVILEGES ON prediction_game.* TO 'pg_migrate'@'localhost';

FLUSH PRIVILEGES;
```

Create the Database

```
CREATE DATABASE IF NOT EXISTS prediction_game
  CHARACTER SET utf8mb4
  COLLATE utf8mb4_unicode_ci;
```

Entity Relationship Diagram

```
erDiagram
    admins {
        int id PK "AUTO_INCREMENT"
        varchar50 username "UNIQUE NOT NULL"
        varchar255 password_hash "NOT NULL"
        timestamp created_at "DEFAULT CURRENT_TIMESTAMP"
```

```

        timestamp    updated_at          "ON UPDATE CURRENT_TIMESTAMP"
    }

    matches {
        int           id                  PK "AUTO_INCREMENT"
        varchar100    team_a              "NOT NULL"
        varchar100    team_b              "NOT NULL"
        varchar150    label               "NULL"
        datetime      match_date          "NULL"
        enum          result               "team_a|team_b|draw|NULL"
        timestamp     created_at           "DEFAULT CURRENT_TIMESTAMP"
        timestamp     updated_at           "ON UPDATE CURRENT_TIMESTAMP"
    }

    users {
        int           id                  PK "AUTO_INCREMENT"
        varchar100    display_name         "UNIQUE NOT NULL"
        timestamp     created_at           "DEFAULT CURRENT_TIMESTAMP"
        timestamp     updated_at           "ON UPDATE CURRENT_TIMESTAMP"
    }

    predictions {
        int           id                  PK "AUTO_INCREMENT"
        int           user_id             FK "NOT NULL"
        int           match_id            FK "NOT NULL"
        enum          prediction           "team_a|team_b|draw NOT NULL"
        timestamp     created_at           "DEFAULT CURRENT_TIMESTAMP"
        timestamp     updated_at           "ON UPDATE CURRENT_TIMESTAMP"
    }

    users    |--o{ predictions : "makes"
    matches  |--o{ predictions : "has"

```

Relationship Summary

```

admins      (standalone – no FK relations)
|
└─ manages → matches  (1 admin manages many matches)
                        |
                        └─ has → predictions (1 match has many predictions)
                                |
                                └─
users  — makes —————→
(1 user makes many predictions; 1 per match enforced by UNIQUE key)

```

Table Definitions

1. admins

Stores the single admin account. The application supports one admin; the table is designed to allow expansion but the UI enforces a single record.

```

CREATE TABLE admins (
  id          INT          NOT NULL AUTO_INCREMENT,
  username    VARCHAR(50)  NOT NULL,
  password_hash VARCHAR(255) NOT NULL,
  created_at  TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at  TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP
              ON UPDATE CURRENT_TIMESTAMP,

  PRIMARY KEY (id),
  UNIQUE KEY uq_admins_username (username)
) ENGINE=InnoDB
  DEFAULT CHARSET=utf8mb4
  COLLATE=utf8mb4_unicode_ci;

```

Column	Type	Nullable	Default	Notes
id	INT	NO	AUTO_INCREMENT	Surrogate primary key
username	VARCHAR(50)	NO	—	Unique login name
password_hash	VARCHAR(255)	NO	—	bcrypt hash (cost ≥ 12)
created_at	TIMESTAMP	NO	CURRENT_TIMESTAMP	Record creation time
updated_at	TIMESTAMP	NO	CURRENT_TIMESTAMP	Auto-updated on write

2. matches

Stores the 10 World Cup match fixtures and their official results. The `result` column is `NULL` until the admin records the outcome.

```

CREATE TABLE matches (
  id          INT          NOT NULL AUTO_INCREMENT,
  team_a      VARCHAR(100) NOT NULL,
  team_b      VARCHAR(100) NOT NULL,
  label       VARCHAR(150)  NULL,
  match_date  DATETIME      NULL,
  result      ENUM(
    'team_a',
    'team_b',
    'draw'
  ) NULL DEFAULT NULL,
  created_at  TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at  TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP
              ON UPDATE CURRENT_TIMESTAMP,

  PRIMARY KEY (id)
) ENGINE=InnoDB
  DEFAULT CHARSET=utf8mb4
  COLLATE=utf8mb4_unicode_ci;

```

Column	Type	Nullable	Default	Notes
id	INT	NO	AUTO_INCREMENT	Surrogate primary key
team_a	VARCHAR(100)	NO	—	First team name
team_b	VARCHAR(100)	NO	—	Second team name
label	VARCHAR(150)	YES	NULL	Display label, e.g. "Group A – Match 1"
match_date	DATETIME	YES	NULL	Scheduled kick-off (UTC)
result	ENUM	YES	NULL	NULL = not yet played
created_at	TIMESTAMP	NO	CURRENT_TIMESTAMP	
updated_at	TIMESTAMP	NO	CURRENT_TIMESTAMP	Auto-updated on write

ENUM values:

Value	Meaning
team_a	Team A wins
team_b	Team B wins
draw	Match drawn
NULL	Result not yet recorded

3. users

Stores participants shown on the leaderboard. Users are created by the admin — there is no self-registration.

```
CREATE TABLE users (  
  id INT NOT NULL AUTO_INCREMENT,  
  display_name VARCHAR(100) NOT NULL,  
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP  
    ON UPDATE CURRENT_TIMESTAMP,  
  
  PRIMARY KEY (id),  
  UNIQUE KEY uq_users_display_name (display_name)  
) ENGINE=InnoDB  
DEFAULT CHARSET=utf8mb4  
COLLATE=utf8mb4_unicode_ci;
```

Column	Type	Nullable	Default	Notes
id	INT	NO	AUTO_INCREMENT	Surrogate primary key

display_name	VARCHAR(100)	NO	—	Public name shown on leaderboard; must be unique
created_at	TIMESTAMP	NO	CURRENT_TIMESTAMP	
updated_at	TIMESTAMP	NO	CURRENT_TIMESTAMP	Auto-updated on write

4. predictions

Stores each user's prediction for each match. A user can have at most one prediction per match (enforced by the composite unique key). Correct/incorrect status is derived at query time by comparing `prediction` to `matches.result`.

```
CREATE TABLE predictions (
  id          INT          NOT NULL AUTO_INCREMENT,
  user_id     INT          NOT NULL,
  match_id    INT          NOT NULL,
  prediction  ENUM(
    'team_a',
    'team_b',
    'draw'
  )          NOT NULL,
  created_at  TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at  TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
              ON UPDATE CURRENT_TIMESTAMP,

  PRIMARY KEY (id),
  UNIQUE KEY uq_predictions_user_match (user_id, match_id),
  CONSTRAINT fk_predictions_user
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE ON UPDATE
    CASCADE,
  CONSTRAINT fk_predictions_match
    FOREIGN KEY (match_id) REFERENCES matches(id) ON DELETE CASCADE ON UPDATE
    CASCADE
) ENGINE=InnoDB
  DEFAULT CHARSET=utf8mb4
  COLLATE=utf8mb4_unicode_ci;
```

Column	Type	Nullable	Default	Notes
id	INT	NO	AUTO_INCREMENT	Surrogate primary key
user_id	INT	NO	—	FK → users.id
match_id	INT	NO	—	FK → matches.id
prediction	ENUM	NO	—	The user's chosen outcome
created_at	TIMESTAMP	NO	CURRENT_TIMESTAMP	
updated_at	TIMESTAMP	NO	CURRENT_TIMESTAMP	Auto-updated on write

Relationships

Relationship	Type	Parent	Child	FK Column	On Delete	On Update
User → Predictions	One-to-Many	users.id	predictions.user_id	user_id	CASCADE	CASCADE
Match → Predictions	One-to-Many	matches.id	predictions.match_id	match_id	CASCADE	CASCADE

Cascade behaviour:

- Deleting a `user` automatically deletes all their `predictions` .
- Deleting a `match` automatically deletes all `predictions` for that match.

Indexes

Table	Index Name	Columns	Type	Purpose
admins	PRIMARY	id	PRIMARY	Row lookup
admins	uq_admins_username	username	UNIQUE	Fast login lookup, enforce uniqueness
matches	PRIMARY	id	PRIMARY	Row lookup
matches	idx_matches_result	result	INDEX	Filter unresolved matches efficiently
users	PRIMARY	id	PRIMARY	Row lookup
users	uq_users_display_name	display_name	UNIQUE	Enforce uniqueness, leaderboard sort
predictions	PRIMARY	id	PRIMARY	Row lookup
predictions	uq_predictions_user_match	user_id, match_id	UNIQUE	One prediction per user per match
predictions	idx_predictions_match_id	match_id	INDEX	Fast join when loading predictions by match

Leaderboard Query (key query — must be fast)

```
SELECT
  u.id,
  u.display_name,
  COUNT(CASE WHEN p.prediction = m.result THEN 1 END) AS total_points,
  RANK() OVER (ORDER BY COUNT(CASE WHEN p.prediction = m.result THEN 1 END) DESC)
```

```
AS `rank`
FROM users u
LEFT JOIN predictions p ON p.user_id = u.id
LEFT JOIN matches m ON m.id = p.match_id
GROUP BY u.id, u.display_name
ORDER BY total_points DESC;
```

This query uses the composite unique index on `predictions(user_id, match_id)` and the primary key on `matches(id)` — no additional indexes needed for up to 100 users and 10 matches.

Constraints & Business Rules

Rule	Enforced By
Max 10 matches	Application layer (check count before INSERT)
One prediction per user per match	UNIQUE KEY <code>uq_predictions_user_match</code>
<code>display_name</code> must be unique	UNIQUE KEY <code>uq_users_display_name</code>
<code>username</code> must be unique	UNIQUE KEY <code>uq_admins_username</code>
Passwords stored as bcrypt hash only	Application layer
<code>result</code> is NULL until admin sets it	Default NULL, set by admin action
Deleting a user removes their predictions	ON DELETE CASCADE on <code>fk_predictions_user</code>
Deleting a match removes its predictions	ON DELETE CASCADE on <code>fk_predictions_match</code>

Migration Strategy

All schema changes are versioned SQL files in `db/migrations/`. Migrations are applied in filename order and must never be edited after being pushed to the shared repository.

File Naming

```
YYYYMMDDHHMMSS_<description>.sql
```

Example:

```
db/migrations/
├─ 20260611000001_create_admins_table.sql
├─ 20260611000002_create_matches_table.sql
├─ 20260611000003_create_users_table.sql
├─ 20260611000004_create_predictions_table.sql
└─ 20260611000005_add_idx_matches_result.sql
```

Migration File Template


```

-- Migration: 20260611000001_create_admins_table.sql
-- Description: Create admins table for admin authentication
-- Author: <author>
-- Date: 2026-06-11

-- ===== UP =====

CREATE TABLE admins (
    ...
);

-- ===== DOWN =====

DROP TABLE IF EXISTS admins;

```

Rules

- Never edit a migration after it has been committed and pushed.
- To undo a schema change, write a new DOWN migration.
- Migrations must be idempotent where possible (CREATE TABLE IF NOT EXISTS , DROP TABLE IF EXISTS).
- Always include a DOWN block for every migration.
- Run migrations as pg_migrate user, not the app user.

Seed Data

A seed file provisions the initial admin account. Run once on first deploy.

File: db/seeds/001_admin_seed.sql

```

-- Replace the hash below with: bcrypt.hashSync('<your-password>', 12)
INSERT INTO admins (username, password_hash)
VALUES ('admin', '$2b$12$REPLACE_WITH_REAL_BCRYPT_HASH')
ON DUPLICATE KEY UPDATE updated_at = CURRENT_TIMESTAMP;

```

Never store a plaintext password in any file. Generate the bcrypt hash locally and paste only the hash into the seed file.

Backup & Maintenance

Backup Command

```

mysqldump \
  --host=127.0.0.1 \
  --user=pg_migrate \
  --password \
  --single-transaction \
  --routines \

```

```
--triggers \
prediction_game > backup_$(date +%Y%m%d_%H%M%S).sql
```

Recommended Practices

Practice	Detail
Automated backups	Daily full dump, retained for 30 days
Point-in-time recovery	Enable MySQL binary logging (binlog) in production
Connection encryption	Use TLS (--ssl-mode=REQUIRED) in production
Password rotation	Rotate pg_app and pg_migrate passwords every 90 days
Slow query log	Enable with long_query_time=1 to catch unindexed queries